

Networks and Systems Management



IP Addressing
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Rationale for IP Addresses



Domain names, IP addresses, MAC addresses

Functional requirements (IP / location vs. MAC / factory)

Usage requirements (names - users, IP / MAC computers)



Global routing vs. private



Embedded addressing in applications



New addressing paradigms



IP Address Formats

IPv4 (32-bit) vs. IPv6 (128-bit)

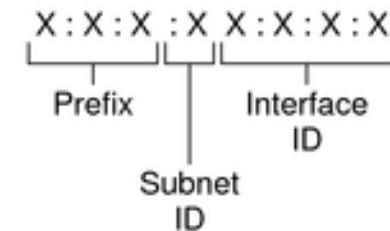
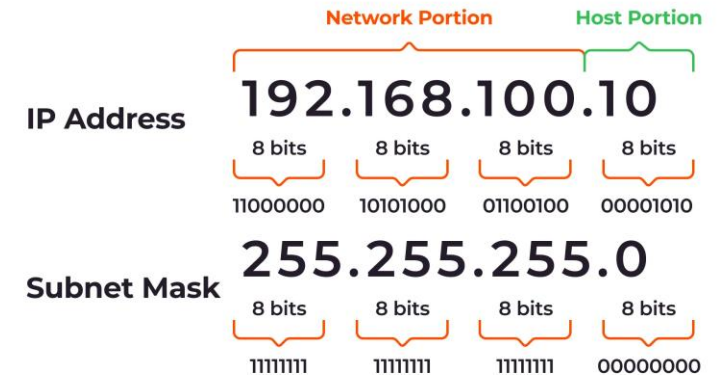
Address structure: Network ID | Interface ID

IPv4 address classes (A/B/C/D/E) vs. CIDR (Classless Inter-Domain Routing)

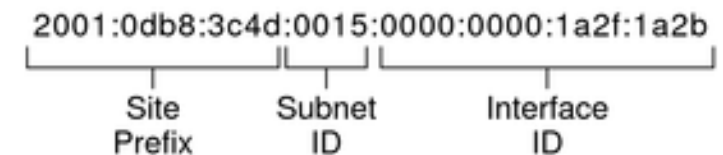
Special ranges (Private, Multicast, Reserved)

Subnetting and Supernetting

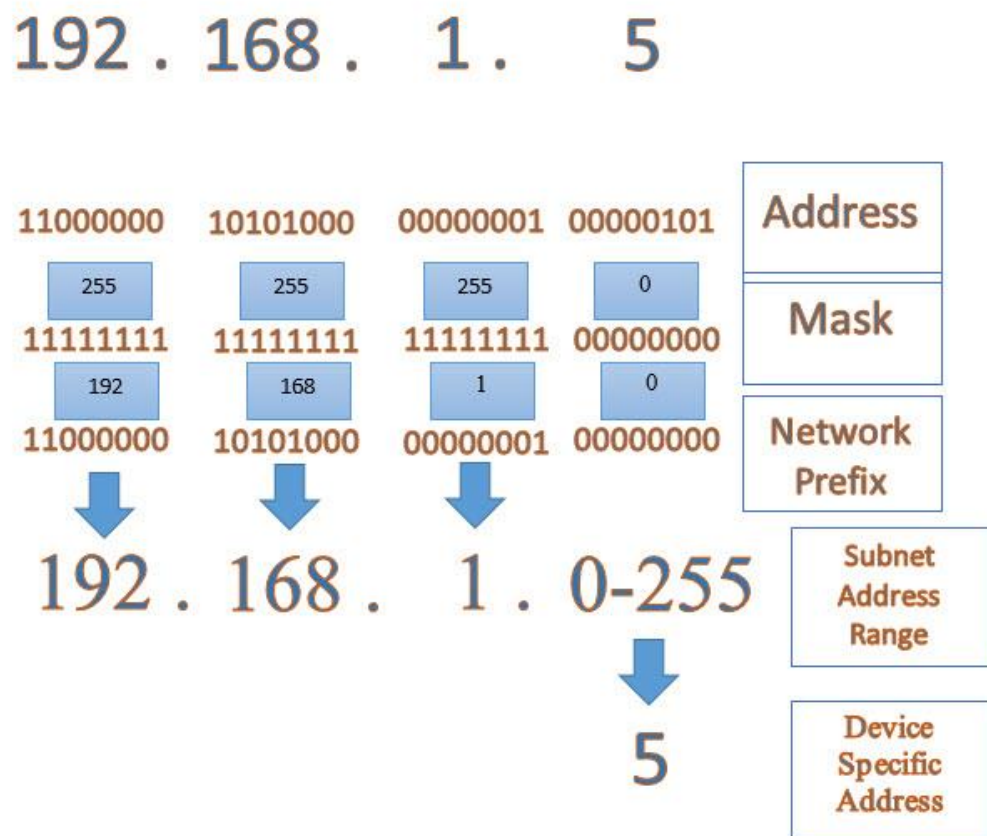
Binary Notation of IP Address and Subnet



Example:



Example Addresses and Network Mask



Address Class	RANGE	Default Subnet Mask
A	1.0.0.0 to 126.255.255.255	255.0.0.0
B	128.0.0.0 to 191.255.255.255	255.255.0.0
C	192.0.0.0 to 223.255.255.255	255.255.255.0
D	224.0.0.0 to 239.255.255.255	Reserved for Multicasting
E	240.0.0.0 to 254.255.255.255	Experimental

Note: Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback testing.

Dimensions of Networking - Special Use Addresses

Network and
Broadcast
address for
192.168.1.0/24

Network address: 192.168.1.0

Broadcast address: 192.168.1.255

Reserved
ranges:

0.0.0.0/8 – This network, this host

127.0.0.0/8 – Loopback

169.254.0.0/16 – Link-local

10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16 – Private/NAT

224.0.0.0/4 – Multicast

FC00::/7 – Unique Local Addresses



Splitting and Aggregating Networks

Subnetting:
Breaking a
network range
into smaller
subnets

Supernetting:
Aggregating
smaller
networks into
a larger one

Example:

- Splitting **200.17.30.0/24** into multiple /27 subnets
- Aggregating /27 subnets into /26

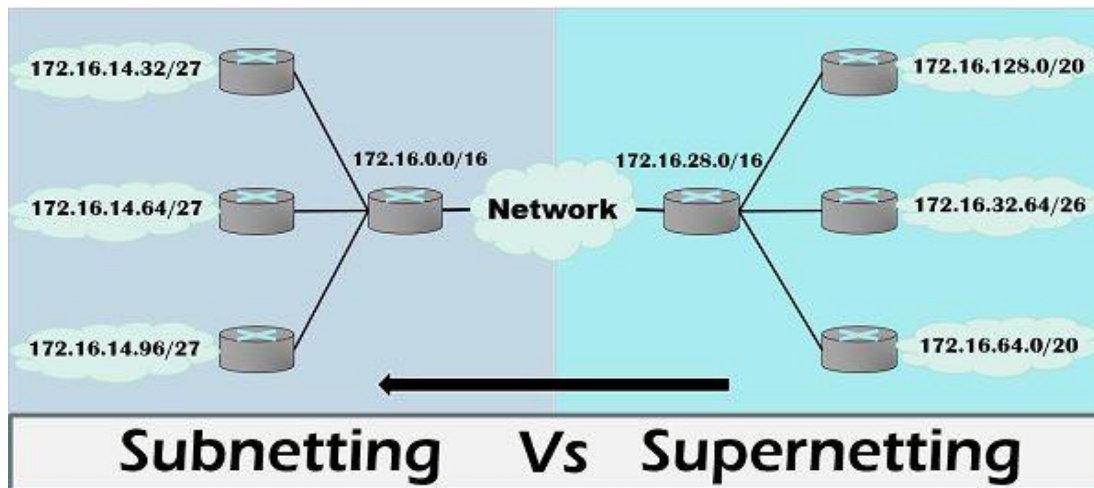
Subnetting and Supernetting



Why subnet? (Large networks split into smaller ones)



Why supernet? (Combine networks into one for efficiency)



Trade-offs (unused addresses, routing efficiency, security)

Subnetting

201.215.128.65/28 Magic = 16

Interesting octet in IP address: **65**

Take Magic and add until we get to **65** without passing

16

32

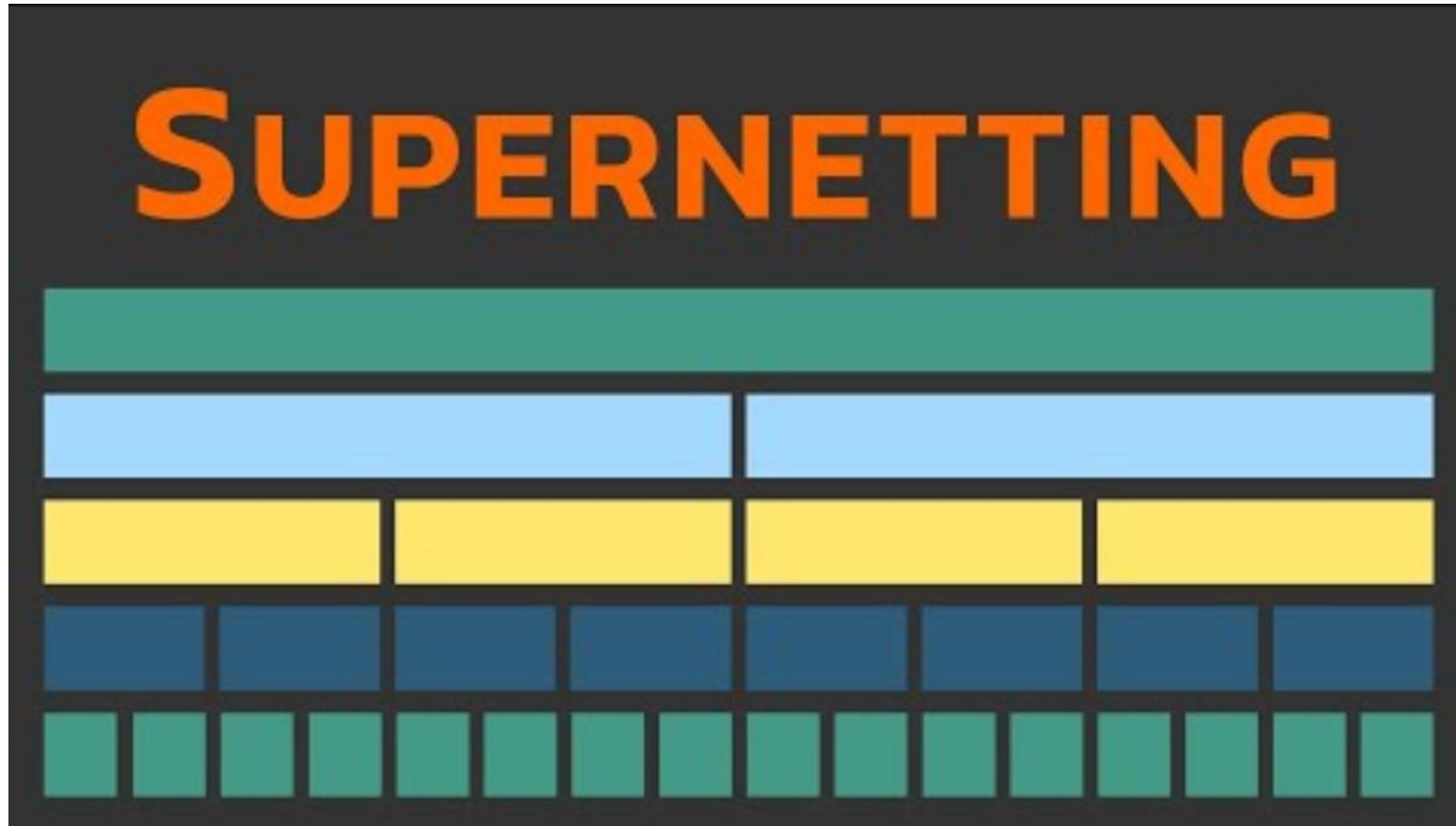
48

64 ---- STAAHP!



<https://www.youtube.com/watch?v=X9jMFufqMgY&t=642s>

Subnetting



<https://www.youtube.com/watch?v=Q4MArJTbUwk>

Key Takeaways

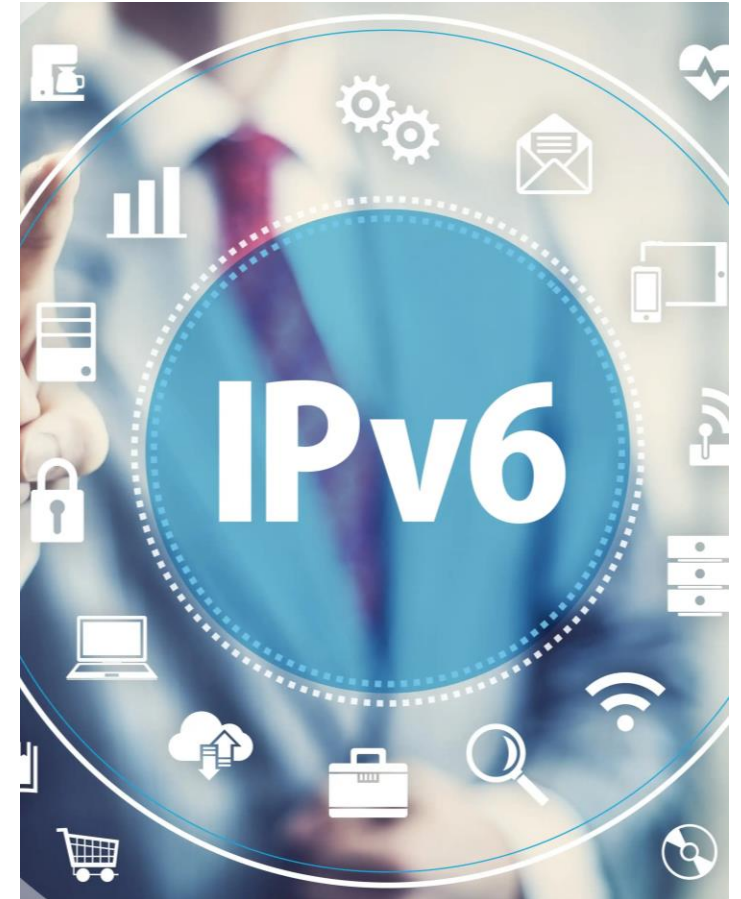
IPv4 addressing has limitations due to address exhaustion, leading to IPv6 adoption.

CIDR provides more efficient IP allocation compared to classful addressing.

Subnetting allows better utilization of IP addresses and improves network segmentation.

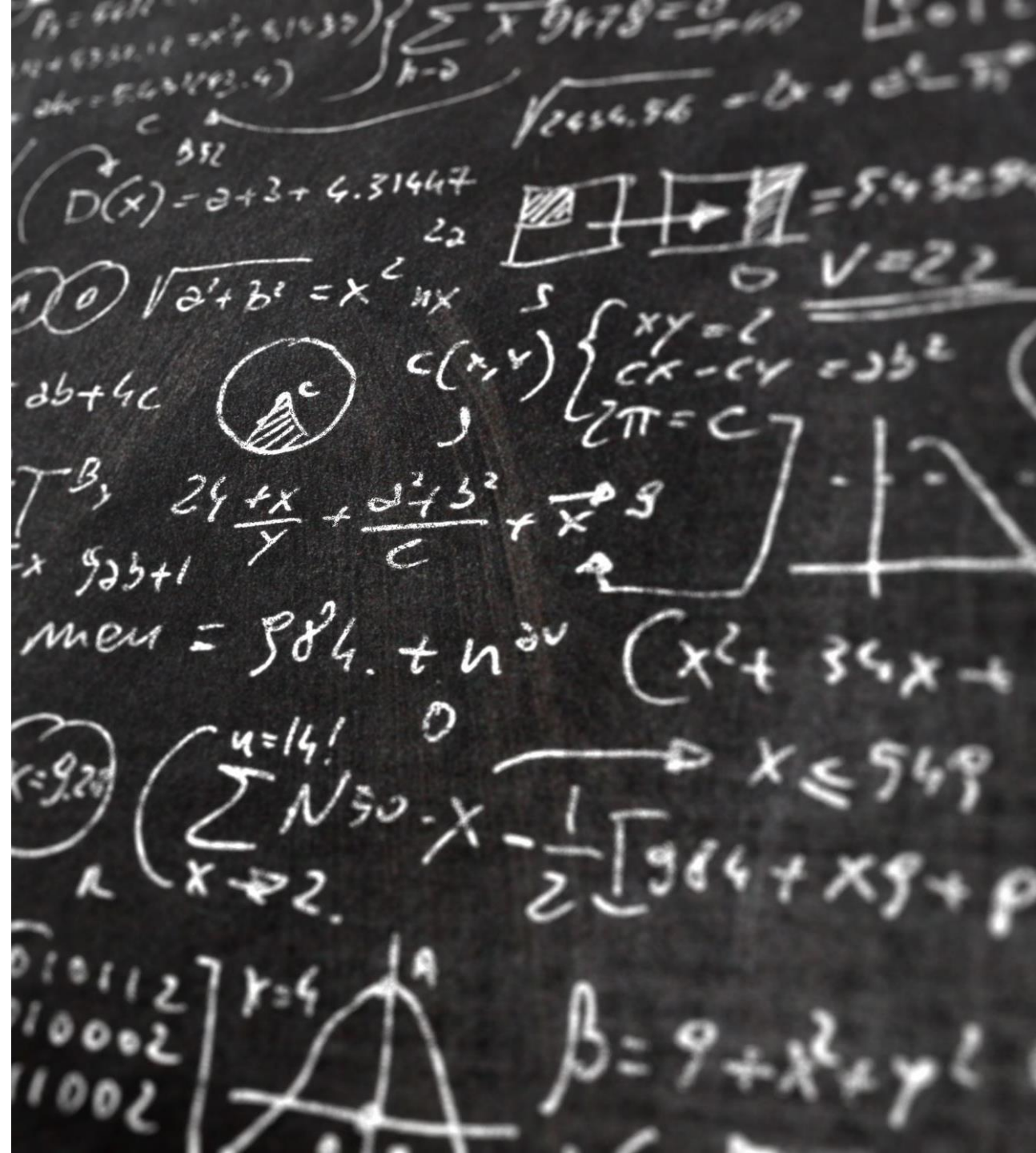
Supernetting reduces routing table size and enhances scalability.

Standards organizations like IETF and IEEE play a key role in defining networking protocols and best practices.



Further Reading & References

- **RFC 791** – IPv4 Specification
- **RFC 2460** – IPv6 Specification
- **RFC 1519** – CIDR
- **Subnet Calculator:** [Subnet Tool](#)
- **Book:** "Computer Networking" by Kurose & Ross



Discussion & Questions

How has the transition from IPv4 to IPv6 impacted modern networking?

What are the advantages and disadvantages of using CIDR over traditional classful addressing?

How do subnetting and supernetting improve network efficiency and scalability?

What role do organizations like IETF and IEEE play in shaping IP addressing standards?



The image shows a detailed IPv4 CIDR chart from the RIPE NCC. It lists all possible IPv4 subnets from /32 down to /0, organized by the number of bits in the prefix. The chart is presented in a table format with columns for IP Addresses, Bits, Prefix, and Subnet Mask. The RIPE NCC logo is visible in the bottom left corner, and contact information for registration services is in the bottom right corner.

IP Addresses	Bits	Prefix	Subnet Mask
1	0	/32	255.255.255.255
2	1	/31	255.255.255.254
4	2	/30	255.255.255.252
8	3	/29	255.255.255.248
16	4	/28	255.255.255.240
32	5	/27	255.255.255.224
64	6	/26	255.255.255.192
128	7	/25	255.255.255.128
256	8	/24	255.255.255.0
512	9	/23	255.255.254.0
1 K	10	/22	255.255.252.0
2 K	11	/21	255.255.248.0
4 K	12	/20	255.255.240.0
8 K	13	/19	255.255.224.0
16 K	14	/18	255.255.192.0
32 K	15	/17	255.255.128.0
64 K	16	/16	255.255.0.0
128 K	17	/15	255.254.0.0
256 K	18	/14	255.252.0.0
512 K	19	/13	255.248.0.0
1 M	20	/12	255.240.0.0
2 M	21	/11	255.224.0.0
4 M	22	/10	255.192.0.0
8 M	23	/9	255.128.0.0
16 M	24	/8	255.0.0.0
32 M	25	/7	254.0.0.0
64 M	26	/6	252.0.0.0
128 M	27	/5	248.0.0.0
256 M	28	/4	240.0.0.0
512 M	29	/3	224.0.0.0
1024 M	30	/2	192.0.0.0
2048 M	31	/1	128.0.0.0
4096 M	32	/0	0.0.0.0